

WEEK 5 REPORT
PREDICTIVE ANALYSIS & SCENARIO SIMULATION

Domain:Apparel & Textiles Industry

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ABSTRACT

This report presents a predictive analysis and scenario simulation plan for the Apparel and Textiles industry using machine learning techniques. The main purpose is to understand how historical sales, seasonal demand, customer behavior, product characteristics, pricing, promotions, and inventory information can be used to predict future trends and support better business decisions. The proposed work uses hypothetical public-domain-style datasets so that the methodology can be designed without depending on confidential company data. The plan considers demand forecasting, customer purchase behavior, and inventory requirements as the main prediction areas. Several scenarios are proposed, including normal demand, festival-season demand, promotional demand, sudden demand reduction, and supply disruption. The simulation will compare predicted demand with inventory availability and identify possible stock-out or overstock situations. Validation will be performed using time-aware train-test splitting, cross-validation where appropriate, and error metrics such as MAE, RMSE, MAPE, and R^2 . Classification metrics can also be used when customer purchase behavior is modeled as a classification problem. The report also discusses data preparation, feature engineering, uncertainty, data leakage, bias, changing consumer preferences, and other risks. The overall approach is designed to be practical, understandable, and suitable for future implementation in an apparel business environment.

1. INTRODUCTION

Predictive analysis is an important application of machine learning because it helps organizations use existing data to estimate what may happen in the future. In the Apparel and Textiles industry, demand can change quickly because of seasons, festivals, fashion trends, discounts, customer preferences, weather, product availability, and economic conditions. Because of this variation, simply looking at previous sales numbers may not always be enough to plan inventory. A predictive model can combine several factors and provide an estimated demand for future periods.

This week's project proposes a structured predictive analysis and scenario simulation framework. The objective is not only to produce a prediction but also to test how the prediction changes when important business conditions change. For example, a retailer may want to know what could happen if festival demand increases, if a discount campaign increases sales, or if supplier delays reduce available stock. Scenario simulation allows these situations to be studied before decisions are implemented.

2. OBJECTIVES

- Develop a predictive analysis plan for Apparel and Textiles applications.
- Identify variables that can influence product demand and customer behavior.
- Formulate practical hypotheses that can be tested using historical or hypothetical public data.
- Design a scenario-based simulation for demand and inventory planning.
- Compare predicted demand with available inventory under different conditions.
- Define suitable validation methods and error measurement metrics.
- Identify risks such as data leakage, missing data, demand volatility, and model drift.
- Suggest mitigation methods that can improve predictive reliability.
- Create an end-to-end workflow that can later be implemented using Python and machine learning libraries.

3. INDUSTRY PROBLEM STATEMENT

Apparel businesses regularly face the problem of deciding how much stock to keep for each product. Too much inventory can increase storage costs and create unsold stock, while too little inventory can lead to stock-outs and missed sales. Demand is also affected by product category, size, color, price, discount, season, festival period, previous sales, customer segment, and marketing activity.

The proposed predictive system will therefore focus on three connected questions: What is likely to be demanded? Which customer or product patterns are changing? How much inventory should be prepared for different possible situations?

4. HYPOTHESIS FORMULATION

SNo.	Hypothesis	Important Variables	Expected Observation
H1	Seasonal and festival periods increase demand for selected apparel categories.	Month, season, festival flag, category, previous sales	Higher predicted demand during relevant periods.
H2	Discounts and promotional campaigns influence purchase quantity.	Discount %, promotion flag, price, category	Demand changes when promotion intensity changes.
H3	Previous sales and recent demand are strong predictors of near-future demand.	Lag sales, rolling average, previous-week sales	Recent demand improves forecast accuracy.
H4	Customer characteristics and past purchase behavior can help predict repeat purchases.	Customer segment, frequency, recency, category history	Some groups show higher repeat-purchase probability.
H5	Inventory availability affects observed sales and can hide true demand.	Opening stock, stock-out days, sales, replenishment	Low stock may cause observed sales to underestimate true demand.

5. PROPOSED DATA SOURCES AND HYPOTHETICAL DATASET

The project can be demonstrated with a hypothetical dataset constructed from variables commonly available in public retail, e-commerce, or machine-learning datasets. Where an actual company dataset is unavailable, a public retail sales dataset can be adapted to an apparel context. No private customer information is required for the proposed simulation.

- Daily or weekly product sales quantity.
- Product category such as shirts, trousers, dresses, jackets, sportswear, or accessories.
- Product price and discount percentage.
- Season and month.
- Festival or promotional-period indicator.
- Opening and closing inventory.
- Number of stock-out days.
- Marketing or campaign indicator.
- Customer segment or anonymized customer group.
- Historical purchase frequency and recent purchase activity.
- Supplier lead time and replenishment quantity.
- Optional external variables such as weather or holiday indicators.

6. DATA DICTIONARY / MAIN VARIABLES

Variable	Type	Example	Purpose
Date	Date	2026-10-15	Time-based forecasting
Product_Category	Categorical	T-Shirts	Product-level demand analysis
Price	Numeric	799	Pricing effect
Discount_Percent	Numeric	20	Promotion effect
Sales_Quantity	Numeric	145	Main prediction target
Opening_Stock	Numeric	300	Inventory simulation
Stockout_Days	Numeric	1	Availability constraint
Season	Categorical	Festival	Seasonal effect
Promotion_Flag	Binary	1	Marketing influence
Customer_Segment	Categorical	Repeat Buyer	Customer behavior
Lead_Time_Days	Numeric	7	Supply planning
Replenishment_Qty	Numeric	250	Inventory decision

7. PREDICTIVE MODELING APPROACH

The predictive analysis can use different machine learning approaches depending on the target variable. Demand quantity is a regression or time-series forecasting problem, while repeat purchase can be treated as a classification problem. Customer or product grouping can be studied using clustering. Model selection should consider accuracy, interpretability, computational requirements, and suitability for the available data.

7.1 Candidate Models

- **Baseline / Moving Average:** a simple reference forecast to determine whether machine learning adds value.
- **Linear Regression:** an interpretable baseline when relationships are approximately linear.
- **Random Forest Regressor:** captures nonlinear relationships among price, promotions, season, category, and historical demand.
- **Gradient Boosting / XGBoost-style boosting:** suitable for structured tabular data and complex interactions.
- **Logistic Regression:** can predict outcomes such as whether a customer is likely to make a repeat purchase.
- **K-Means Clustering:** can group customers or products according to behavioral or sales characteristics.

8. FEATURE ENGINEERING

Feature engineering converts raw information into variables that are more useful for prediction. Proposed features include lagged sales, rolling averages, month, week number, season, festival indicators, discount level, price changes, promotion status, stock availability, and supplier lead time.

- Sales_Lag_1: sales from the previous period.
- Sales_Lag_7: comparable previous-week sales for daily data.
- Rolling_Mean_7: average sales over the previous seven periods.
- Rolling_Mean_30: average sales over the previous thirty periods.
- Price_Change: percentage change from the previous price.
- Promotion_Flag: whether a campaign is active.
- Festival_Flag: whether a major festival or seasonal event is active.
- Stock_Availability_Rate: proportion of the period during which the product was available.
- Lead_Time_Risk: an indicator derived from supplier lead time.

9. SIMULATION DESIGN

The scenario simulation will create several hypothetical business conditions and estimate their effect on demand and inventory. The simulation can cover approximately 12 months of daily or weekly observations so that seasonal patterns can be represented. Each scenario uses the same underlying product and customer structure while changing selected input variables.

9.1 Baseline Scenario

The baseline represents normal operating conditions using typical price, promotion frequency, supplier lead time, and seasonal demand. It becomes the reference point for comparison.

9.2 Scenario A – Festival Demand Increase

Demand for selected categories is increased during a festival period. The simulation checks whether current inventory is sufficient and identifies products that may need additional replenishment.

9.3 Scenario B – Promotional Campaign

A discount or promotional campaign is activated. The simulation studies whether the expected increase in sales justifies additional inventory and possible margin impact.

9.4 Scenario C – Demand Reduction

Demand decreases because of weak market interest or changing preferences. The simulation checks the possibility of excess inventory and more conservative replenishment.

9.5 Scenario D – Supply Disruption

Supplier lead time increases or replenishment quantity falls. The simulation estimates potential stock-outs and safety-stock requirements.

9.6 Scenario E – Combined Stress Test

Demand rises while supplier availability becomes limited. This tests the robustness of the inventory plan under difficult conditions.

10. SIMULATION VARIABLES AND DURATION

Component	Proposed Setting	Reason	Output
Duration	12 months	Captures seasonal variation	Monthly/weekly forecasts
Time unit	Daily or weekly	Depends on data availability	Demand time series
Products	Multiple categories	Allows category comparison	Category-level demand
Scenario change	$\pm 10\%$ to $\pm 30\%$ for selected factors	Stress testing	Scenario impact
Inventory	Opening stock + replenishment	Tests stock planning	Closing stock
Supply	Normal and delayed lead time	Tests disruption risk	Stock-out risk
Promotion	Normal vs campaign	Tests marketing effect	Incremental demand

11. EXPECTED OUTPUTS

- Predicted demand for each product/category and future period.
- Expected inventory balance after predicted sales.
- Estimated stock-out quantity or number of stock-out periods.
- Estimated excess inventory under low-demand scenarios.
- Comparison of baseline and alternative scenarios.
- Identification of products requiring additional safety stock.
- Customer groups with higher predicted repeat-purchase probability.
- Error measurements showing prediction reliability.

12. VALIDATION STRATEGY

Validation is essential because a prediction that performs well on training data may perform poorly on future data. Since demand is time-dependent, the project should avoid randomly mixing future observations into the training set. A chronological split is preferred for the main forecasting evaluation.

12.1 Time-Based Train-Test Split

For example, the first 70% of chronological observations can be used for training, the next 15% for validation, and the final 15% for testing. The exact percentages can be adjusted according to dataset size. The important principle is that future information must not be used to train the model before it is predicted.

12.2 Cross-Validation

Time-series cross-validation, also called walk-forward validation, can be used for forecasting. The model is repeatedly trained on an earlier time window and evaluated on the next time window. For non-time-dependent classification experiments, k-fold cross-validation can be considered if it does not create leakage.

13. ERROR MEASUREMENT METRICS

Metric	Use	Interpretation	Preferred Direction
MAE	Demand forecasting	Average absolute prediction error	Lower
RMSE	Demand forecasting	Penalizes larger errors more strongly	Lower
MAPE	Forecast comparison	Average percentage error; caution near zero actual values	Lower
R ²	Regression	Explains proportion of target variation	Higher
Accuracy	Classification	Correct predictions / total predictions	Higher
Precision	Classification	Reliability of positive predictions	Higher
Recall	Classification	Ability to identify actual positive cases	Higher
F1-score	Classification	Balance between precision and recall	Higher

14. SCENARIO COMPARISON METHOD

After generating predictions, each scenario will be compared against the baseline. The analysis will calculate percentage change in predicted demand, inventory requirement, stock-out risk, and excess stock. This makes the simulation useful for decision support rather than treating the model as one fixed answer.

- Demand impact = scenario predicted demand compared with baseline demand.
- Inventory gap = available inventory minus predicted demand.
- Stock-out flag = activated when predicted demand exceeds available inventory.
- Safety stock requirement = additional quantity needed to maintain a chosen service level.
- Excess stock indicator = inventory remaining after expected demand beyond a predefined threshold.

15. RISK ASSESSMENT

Risk	Possible Effect	Mitigation	Monitoring
Poor-quality data	Unreliable predictions	Data cleaning and validation rules	Missing/outlier reports
Data leakage	Artificially high test performance	Time-aware splitting and feature review	Leakage checks
Sudden fashion trend	Forecast error increases	Frequent retraining and external signals	Error drift
Stock-out distortion	True demand underestimated	Include availability variables	Stock-out analysis
Seasonal change	Wrong seasonal assumptions	Recent rolling validation	Seasonal error tracking
Model overfitting	Poor future performance	Regularization, validation, simple baselines	Train-test gap
Supplier disruption	Inventory shortage	Safety stock and scenario testing	Lead-time monitoring
Customer behavior change	Repeat-purchase prediction falls	Periodic model updates	Segment performance
Class imbalance	Poor minority-class detection	Precision/recall/F1 and suitable sampling	Class-wise metrics

16. RISK MITIGATION PLAN

- Perform data-quality checks before model training.
- Keep training, validation, and testing periods strictly separated for time-based problems.
- Use simple baseline forecasts before adopting complex models.
- Monitor errors separately for product categories and seasons.
- Retrain the model when demand patterns change significantly.
- Use scenario simulation instead of relying on one forecast.
- Keep human review in the final inventory decision process.
- Document model assumptions and the meaning of each feature.
- Maintain a record of model versions, datasets, and evaluation results.

17. PROPOSED END-TO-END WORKFLOW

Data Collection

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Data Cleaning

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Exploratory Analysis

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Feature Engineering

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Hypothesis Testing

→

Baseline Forecast

→

Model Training

→

Time-Based Validation

→

Model Comparison

→

Scenario Generation

→

Demand Prediction

→

Inventory Simulation

→

Risk Analysis

→

Report and Recommendations

18. PRACTICAL EXAMPLE OF THE SIMULATION

Assume a retailer has 500 units of a selected T-shirt category at the beginning of a festival period. The baseline model predicts demand of 420 units. In a festival-demand scenario, the simulation increases relevant demand drivers and the model predicts 510 units. The simulation would then show a possible inventory gap of approximately 10 units before considering additional replenishment or safety stock. If a supplier delay is also introduced, the gap could become larger. These values are illustrative and are not claimed to be actual industry measurements.

19. MODEL EVALUATION PLAN

The final model should not be selected only because it produces the lowest single error value. Evaluation should consider forecast accuracy, stability across time periods, interpretability, training time, scalability, and ease of maintenance. A model that performs consistently across several validation windows may be more useful than one that performs extremely well on only one period.

- Compare the best machine learning model against a simple baseline.
- Check MAE and RMSE for overall forecasting error.
- Review MAPE carefully when actual demand can be zero or very small.
- Compare performance by product category and season.
- Inspect prediction errors using actual-versus-predicted plots.
- Evaluate scenario stability by testing several reasonable input changes.
- Document assumptions and limitations alongside numerical results.

20. TOOLS AND TECHNOLOGIES

- Python for data processing, simulation, and model development.
- Pandas and NumPy for data preparation and numerical operations.
- Matplotlib for visualization of trends, errors, and scenario comparisons.
- Scikit-learn for regression, classification, clustering, validation, and metrics.
- Jupyter Notebook or Google Colab for experimentation and documentation.
- Excel for basic scenario tables and manual validation checks.

21. EXPECTED BENEFITS

- Better understanding of future demand patterns.
- Improved inventory planning.
- Earlier identification of possible stock-outs.
- Reduced risk of excessive inventory.
- Improved evaluation of promotional strategies.
- More structured customer behavior analysis.
- Ability to test business conditions before implementation.
- Clearer understanding of model uncertainty and risk.

22. LIMITATIONS

The proposed project is a planning and simulation framework rather than a production forecasting system. Hypothetical data cannot fully represent the complexity of a real apparel company. Real-world demand may also be affected by sudden fashion trends, competitor actions, supply chain events, weather, or changes in customer preferences. Therefore, simulated results should be treated as analytical examples. A real deployment would require reliable historical data, continuous monitoring, business validation, and periodic model retraining.

CONCLUSION

This Week 5 project develops a structured plan for predictive analysis and scenario simulation in the Apparel and Textiles industry. The proposed framework connects machine learning prediction with practical inventory and customer-behavior questions. By combining historical demand features, seasonal indicators, pricing, promotions, inventory information, and customer behavior variables, the system can estimate future trends and test alternative business situations. The scenario approach is especially useful because apparel demand is not constant and can change with festivals, promotions, supply constraints, and changing consumer preferences. Time-based validation, suitable error metrics, baseline comparisons, and risk monitoring are included to make the analysis more reliable. The framework can later be implemented with a public dataset or company dataset and extended into a complete forecasting and decision-support application.